

| <b>(d) Electromagnetic induction</b>                                                                                                                                                                                                                                       |                                                                                                                                                                                                                 |
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| <b>Students should:</b>                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                 |
| 6.15                                                                                                                                                                                                                                                                       | know that a voltage is induced in a conductor or a coil when it moves through a magnetic field or when a magnetic field changes through it and describe the factors that affect the size of the induced voltage |
| 6.16                                                                                                                                                                                                                                                                       | describe the generation of electricity by the rotation of a magnet within a coil of wire and of a coil of wire within a magnetic field, and describe the factors that affect the size of the induced voltage    |
| <b>6.17P describe the structure of a transformer and understand that a transformer changes the size of an alternating voltage by having different numbers of turns on the input and output sides</b>                                                                       |                                                                                                                                                                                                                 |
| <b>6.18P explain the use of step-up and step-down transformers in the large-scale generation and transmission of electrical energy</b>                                                                                                                                     |                                                                                                                                                                                                                 |
| <b>6.19P know and use the relationship between input (primary) and output (secondary) voltages and the turns ratio for a transformer:</b> $\frac{\text{input (primary) voltage}}{\text{output (secondary) voltage}} = \frac{\text{primary turns}}{\text{secondary turns}}$ |                                                                                                                                                                                                                 |
| <b>6.20P know and use the relationship:</b><br><b>input power = output power</b><br>$V_p I_p = V_s I_s$ <b>for 100% efficiency</b>                                                                                                                                         |                                                                                                                                                                                                                 |